

Prefix Maxima and Consecutive Blocks: An Exact Deterministic Equivalence

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Abstract

For every finite sequence, the largest consecutive-block sum is at most twice the largest prefix sum, while every prefix is itself a consecutive block. Thus a uniform block theorem and the desired prefix theorem are equivalent up to the literal factor two, with no phase, scale, support, or power change. The classification is REDUCTION_L1; no L2 arithmetic progress or endpoint credit is claimed.

1 Frozen target

The selected route is 0161.bad_endpoint_pointwise_fixed_atom [1]. The required signature is actual fixed- h_0 packet, named fixed atom, deterministic every prefix, deterministic every scale, fixed- X power at that atom, and actual active support. Throughout $h_0 = 2$ is only a source-backed data fact.

2 Exact result

Theorem 1. *If $P = \max_k \sum_{j \leq k} a_j$ and $B = \max_r \sum_{r \leq j \leq sa_j} a_j$, then $P \leq B \leq 2P$. The factor two is sharp for general complex sequences.*

Proof. Write $S_k = \sum_{j \leq k} a_j$ and $S_0 = 0$. Every prefix is the block $S_k - S_0$, hence $P \leq B$. Every block is $S_s - S_r$, so $|S_s - S_r| \leq |S_s| + |S_r| \leq 2P$. Prefix values 1 and -1 show that the factor two is sharp for unrestricted complex sequences. $\square$

The smallest missing literal theorem is NAMED_ATOM_CONSECUTIVE_BLOCK_POWER_SAVING.

3 Scoped stop and claim firewall

No new scoped stop is declared in this paper.

The result is L0/L1 only. Phase averages and Lebesgue-almost-every phase are not named atoms. Fixed $h_0 = 2$ is not decay. Archive addressing is not canonical physical representation, and empty-domain truth is not actual active support. The named-atom endpoint charge is zero, so the strict 1/400 budget is unpaid. We claim no program-positive L2, prime-pair lower bound, or twin-prime theorem.

4 Reproducibility

The adjacent JSON payload records the full six-axis signature, source lock, typed result, exact scoped-stop cell, endpoint ledger, and claim firewall. The audit artifact contains eight true mutation-rejection assertions.

References

- [1] Liang Wang. Mvp8 source-phase route decision, 2026.